

1. A group of 150 college students were asked to indicate their most liked film star from six different well known film actors namely A, B, C, D, E and F in order to ascertain their popularity. The observed data were as follows

Actors	A	B	C	D	E	F
Frequencies	24	20	32	25	28	21

Test at 5% whether all actors are equally popular.

Soln:

Actors	Freq (O)	Exp _e (E)	(O-E)	(O-E) ²	(O-E) ² /E
A	24	25	-1	1	0.04
B	20	25	-5	25	1
C	32	25	7	49	1.96
D	25	25	0	0	0
E	28	25	3	9	0.36
F	21	25	-4	16	0.64
Sum	150				4.00

Hypothesis : H_0 : All six actors are equally popular.

$$E = \frac{\text{Total freq}}{\text{no. of actors}} = \frac{150}{6} = 25$$

degree of freedom $df = (n-1)$

$$= 6-1$$

$$df = 5$$

From the table, $t_{(0.005, 5)} = \cancel{11.070} \quad 11.070.$

Since $\chi^2_{cal} < \chi^2_{table} \Rightarrow$ Accept null hypothesis

Inference:

There is no significant difference in the popularity of six film actors. Hence all actors are equally popular.

Latin square Design

According to column, Row & variation are arranged in square is known as Latin square. It is very popular.

The n treatments are given to these plots such that each treatment occurs only once in each row and only in each column. The various possible arrangements obtained in this manner are known as Latin square of order n . This design of experiment is called Latin square Design (LSD).

1) An oil company was interested in testing four different blends of gasoline for fuel efficiency by controlling the variability of four different drivers and four models of cars. The fuel efficiency was measured as kilometers per litre after driving the cars over a standard course.

Analyze the data and write the inference based on the results (given $F(0.05) = 4.76$ for $(3, 6) df$)

A (13)	D (9)	C (10)	B (12)
B (12)	C (10)	A (13)	D (8)
C (10)	B (12)	D (9)	A (13)
D (9)	A (14)	B (12)	C (10.5)

Given:

H_0 :

H_1 :

variance : Drivers, cars, blends of gasoline (treatment)

Total $N = 4 \times 4 = 16$
(D) (C).

Reframe the table.

				total	Treatment
A	D	C	B		
(13)	(9)	(10)	(12)	44	A - 53
B	C	A	D	43	B - 48
(12)	(10)	(13)	(8)		C - 40.5
C	B	D	A	44	D - 35
(10)	(12)	(9)	(13)		
D	A	B	C	45.5	<u>176.5</u>
(9)	(14)	(12)	(10.5)		
Total	44	45	44	43.5	176.5

calculate correction factor (CF)

$$CF = \frac{(\sum X_{total})^2}{N}$$

$$= \frac{(176.5)^2}{16}$$

$$CF = 1947.01$$

calculate sum of squares (SS):

$$SS_{Row} = \frac{44^2}{4} + \frac{43^2}{4} + \frac{44^2}{4} + \frac{45.5^2}{4} - 1947.01$$

$$= 1947.81 - 1947.01$$

$$SS_{row} = 0.7968$$

$$SS_{col} = \frac{44^2 + 46^2 + 44^2 + 43.5^2}{4} - 1947.01$$

$$SS_{col} = 0.2968$$

$$SS_{b/w \text{ treatments}} = \frac{53^2 + 48^2 + 40.5^2 + 35^2}{4} - CF$$

$$SS_{treat} = 47.546$$

$$SS_{error} = SS_{total} - SS_{row} - SS_{col} - SS_{treat}$$

$$SS_{total} = \sum X_{total}^2 - CF$$

$$= 1996.25 - 1947.01$$

$$SS_{total} = 49.234$$

$$SS_{err} = 49.234 - 0.7968 - 0.2968 - 47.546$$

$$SS_{err} = 0.61$$

ANALYSIS

Source of Variation

calculate degree of freedom (df)

$$df_{row} = (4-1) = 3$$

$$df_{col} = (4-1) = 3$$

$$df_{treat} = (4-1) = 3$$

$$df_{error} = (N-1) - (df_{row} + df_{col} + df_{treat})$$

$$= 15 - 9$$

$$df_{err} = 6$$

calculate Mean sum of squares MS

$$MS_{\text{row}} = \frac{SS_{\text{row}}}{df_{\text{row}}} = \frac{0.7968}{3} = 0.2656$$

$$MS_{\text{col}} = \frac{SS_{\text{col}}}{df_{\text{col}}} = \frac{0.2968}{3} = 0.0989$$

$$MS_{\text{treat}} = \frac{SS_{\text{treat}}}{df_{\text{treat}}} = \frac{47.546}{3} = 15.848$$

$$MS_{\text{err}} = \frac{SS_{\text{err}}}{df_{\text{err}}} = \frac{0.61}{6} = 0.1016$$

calculate F factor:

$$F_{\text{row}} = \frac{MS_{\text{row}}}{MS_{\text{err}}} = \frac{0.2656}{0.1016} = 2.61$$

$$F_{\text{col}} = \frac{MS_{\text{col}}}{MS_{\text{err}}} = \frac{0.0989}{0.1016} = 0.9734$$

$$F_{\text{treat}} = \frac{MS_{\text{treat}}}{MS_{\text{err}}} = \frac{15.848}{0.1016} = 155.984$$

ANOVA

source of variance	df	SS	MS	F_{cal}	F_{table}
blw Rows	3	0.7968	0.2656	2.61	4.76
blw columns	3	0.2968	0.0989	0.9734	4.76
blw Treatments	3	47.546	15.848	155.984	4.76
Error	6	0.61	0.1016		

Inference :

- (i) since $F_{cal} < F_{table}$. So we infer that there is no significant difference blw rows. (ie) Accept null hypothesis
(2.61) < (4.76)
- (ii) since $F_{cal} < F_{table}$. so we infer that there is no significant difference blw columns (ie) Accept null hypothesis
(0.9734) < (4.76)
- (iii) since $F_{cal} > F_{table}$. we infer that there is significant difference blw treatments. (ie) Reject null hypothesis and accept alternate hypothesis.
(155.984) > (4.76)

2. The surface finish of products produced in a machine shop is suspected to be affected by factors operator and machine. The data of this experiment with two replications in different treatment combinations are summarized in the given table. Analyze and check the significance of components of the related model when significance level is 0.05.

		Machine (B)	
		1	2
operator (A)	1	65 70	20 40
	2	30 35	50 40

Soln :

H_0 : Product produced in a machine shop is not affected by factors operator and Machine

H_1 : Product produced in a machine shop is ~~not~~ affected by factors operator and Machine.

Soln :

Total $N = 8$, factor = 2

Levels = 2

Interactions = $2^2 = 4$.

1

2 (B)

$$1 \quad \left. \begin{array}{l} 65 \\ 70 \end{array} \right\} 135 \quad \left. \begin{array}{l} 20 \\ 40 \end{array} \right\} 60 \Rightarrow 195$$

(A)

$$2 \quad \left. \begin{array}{l} 30 \\ 35 \end{array} \right\} 65 \quad \left. \begin{array}{l} 50 \\ 40 \end{array} \right\} 90 \Rightarrow 155$$

$$200$$

$$150$$

calculate correction factor (CF)

$$CF = \frac{(\sum X_{total})^2}{N}$$

$$= \frac{(350)^2}{8}$$

$$\boxed{CF = 15312.5}$$

calculate sum of squares (SS)

$$SS_{total} = \sum X_{total}^2 - CF$$

$$= 17350 - 15312.5$$

$$\boxed{SS_{total} = 2037.5}$$

$$SS_A = \frac{\sum A_1^2}{4} + \frac{\sum A_2^2}{4} - CF$$

$$= \frac{195^2}{4} + \frac{155^2}{4} - 15312.5$$

$$\boxed{SS_A = 200}$$

$$SS_B = \frac{\sum B_1^2}{4} + \frac{\sum B_2^2}{4} - CF$$

$$= \frac{200^2}{4} + \frac{150^2}{4} - 15312.5$$

$$SS_B = 312.5$$

$$SS_{AB(\text{subtotal})} = \frac{\sum (A_1 B_1)^2}{2} + \frac{\sum (A_1 B_2)^2}{2} + \frac{\sum (A_2 B_1)^2}{2} + \frac{\sum (A_2 B_2)^2}{2} - (p - CF)$$

$$= \frac{135^2 + 65^2 + 60^2 + 90^2}{2} - 15312.5$$

$$SS_{AB(\text{subtotal})} = 1762.5$$

$$SS_{AB} = SS_{AB(\text{subtotal})} - SS_A - SS_B$$

$$= 1762.5 - 200 - 312.5$$

$$SS_{AB} = 1250$$

Calculate degree of freedom (df)

$$df_A = (2-1) = 1$$

$$df_B = (2-1) = 1$$

$$df_{AB} = (2-1)(2-1) = 1$$

$$df_{\text{error}} = (N-1) - df_A - df_B - df_{AB} = 7 - (3) = 4$$

$$df_{\text{total}} = N-1 = 8-1 = 7$$

$$SS_{\text{error}} = SS_{\text{total}} - SS_A - SS_B - SS_{AB}$$

$$= 2037.5 - 200 - 312.5 - 1250$$

$$SS_{\text{error}} = 275$$

calculate Mean sum of square (MS)

$$MS = \frac{SS}{df}$$

$$MS_A = \frac{SS_A}{df_A} = \frac{200}{1} = 200$$

$$MS_B = \frac{SS_B}{df_B} = \frac{312.5}{1} = 312.5$$

$$MS_{AB} = \frac{SS_{AB}}{df_{AB}} = \frac{1250}{1} = 1250$$

$$MS_{\text{error}} = \frac{SS_{\text{error}}}{df_{\text{error}}} = \frac{275}{4} = 68.75$$

$$MS_{\text{total}} = \frac{SS_{\text{total}}}{df_{\text{total}}} = \frac{2037.5}{7} = 291.07$$

calculate F factor:

$$F_A = \frac{MS_A}{MS_{\text{error}}} = \frac{200}{68.75} = 2.909$$

$$F_B = \frac{MS_B}{MS_{\text{error}}} = \frac{312.5}{68.75} = 4.545$$

$$F_{AB} = \frac{MS_{AB}}{MS_{\text{error}}} = \frac{1250}{68.75} = 18.1818$$

ANOVA Table

Source of variation	df	SS	MS	F_{cal}	$F_{table} (\alpha=0.05)$
A	1	200	200	2.909	7.71
B	1	312.5	312.5	4.545	7.71
AB	1	1250	1250	18.1818	7.71
Exp Error	4	275	68.75		
Total	7	2037.5			

Inference :

(i) since $F_{cal} < F_{table} \rightarrow$ There is no significant effect on
 $2.909 < 7.71$

product by operator and machine $\rightarrow$ ~~that~~ H_0 is accepted

(ii) since $F_{cal} < F_{table} \rightarrow$ There is no significant effect on
(4.545) (7.71)

Product by operator and machine $\rightarrow H_0$ is accepted

(iii) since $F_{cal} > F_{table} \rightarrow$ There is a significant effect on
(18.1818) (7.71)

product by operator and machine $\rightarrow H_1$ is accepted.

4. A factory produces items with a defect rate of 2%. If a sample of 50 items is selected, calculate the probability of finding exactly 3 defective items.

Given data:

no. of samples = 50

Defect rate = 2% = 0.02

to find

$$P(X=3) = ?$$

Soln:

Binomial Distribution is applied.

$$P(X=x) = {}^nC_x p^x q^{n-x}$$

$$P(X=3) = {}^{50}C_3 p^3 q^{50-3}$$

$$= {}^{50}C_3 \times (0.02)^3 \times (1-0.02)^{47}$$

$$= \frac{{}^{25}50 \times {}^{16}49 \times {}^{16}48}{1 \times 2 \times 3} \times 8 \times 10^{-6} \times 0.3869$$

$$\boxed{P(X=3) = 0.0606}$$

5. The number of customer arrivals at service desk follows a distribution with a mean of 5 per hour. what is the probability that exactly 7 customer will arrive in the next hour?

Given Data :

Mean = 5 per hour

To find :

$$P(X=7) = ?$$

Soln:

Poisson Distribution is applied.

$$\therefore \lambda = 5$$

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$P(X=7) = \frac{e^{-5} \times 5^7}{7!}$$

$$= \frac{6.737 \times 10^{-3} \times 78125}{5040}$$

$$P(X=7) = 0.1044$$

6. A clinical trial involves 100 patients where the probability of side effects from a medication is 0.01. using an appropriate distribution, estimate the probability that more than 2 patients will experience side effects. Justify choice of distribution.

Given :

no. of patients $n=100$

probability of side effect $p=0.01$

To find

$$P(X > 2) = ?$$

Soln :

Binomial Distribution is applied.

Justification : The no. of Trails (i.e) no. of patients are finite no. (i.e) 100.

$$P(X=x) = {}^n C_x p^x q^{n-x}$$

$$P(X > 2) = 1 - \sum_{x=0}^2 {}^n C_x p^x q^{n-x}$$

$$= 1 - \left[{}^{100}C_0 p^0 q^{100-0} + {}^{100}C_1 p^1 q^{100-1} + {}^{100}C_2 p^2 q^{100-2} + {}^{100}C_3 p^3 q^{100-3} \right]$$

$$= 1 - (1 + 0.99) - \frac{100!}{10 \times 9} (0.01)(0.99)^9 -$$

$$\frac{100!}{10 \times 9 \times 8} (0.01)^2 (0.99)^8 + \frac{100!}{10 \times 9 \times 8 \times 7} (0.01)^3 (0.99)^7$$

$$= 1 - (0.3660 + 0.3697 + 0.1848) = 0.0795$$

$$P(X > 2) = 0.0795$$

7. A company claims that the mean time of its batteries is 100 hours. A sample of 25 batteries has a mean lifetime of 98 hrs with a standard deviation of 5 hrs. Perform a one sample T-test to evaluate the company's claim at a 5% of significant level.

Given Data :

Population Mean $\mu = 100$ hrs

no. of samples $n = 25$

Sample mean $\bar{X} = 98$ hrs

Sample std deviation $s = 5$ hrs.

Significant level $\alpha = 5\%$

To find

one sample T-test =? @ $\alpha = 0.05$.

Soln :

$$T\text{-test} = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

$$= \frac{98 - 100}{5/\sqrt{25}}$$

$$\boxed{T\text{-test} = -2.00}$$

degree of freedom $df = N - 1 = 25 - 1 = 24$.

for T-test $\frac{5\%}{2} = 2.5\%$ level of significance is considered.

from the table $T\text{-test value} \rightarrow \text{critical } t_{(0.025, 24)} = 2.064.$

$$\text{Here } T\text{-test}_{\text{cal}} < T\text{-test}_{\text{critical}} \\ (-2.00) < (2.064)$$

Since $T\text{-test}_{\text{cal}} < T\text{-test}_{\text{critical}}$. So null hypothesis is accepted.

There is no significant difference between mean life time of batteries and hours.